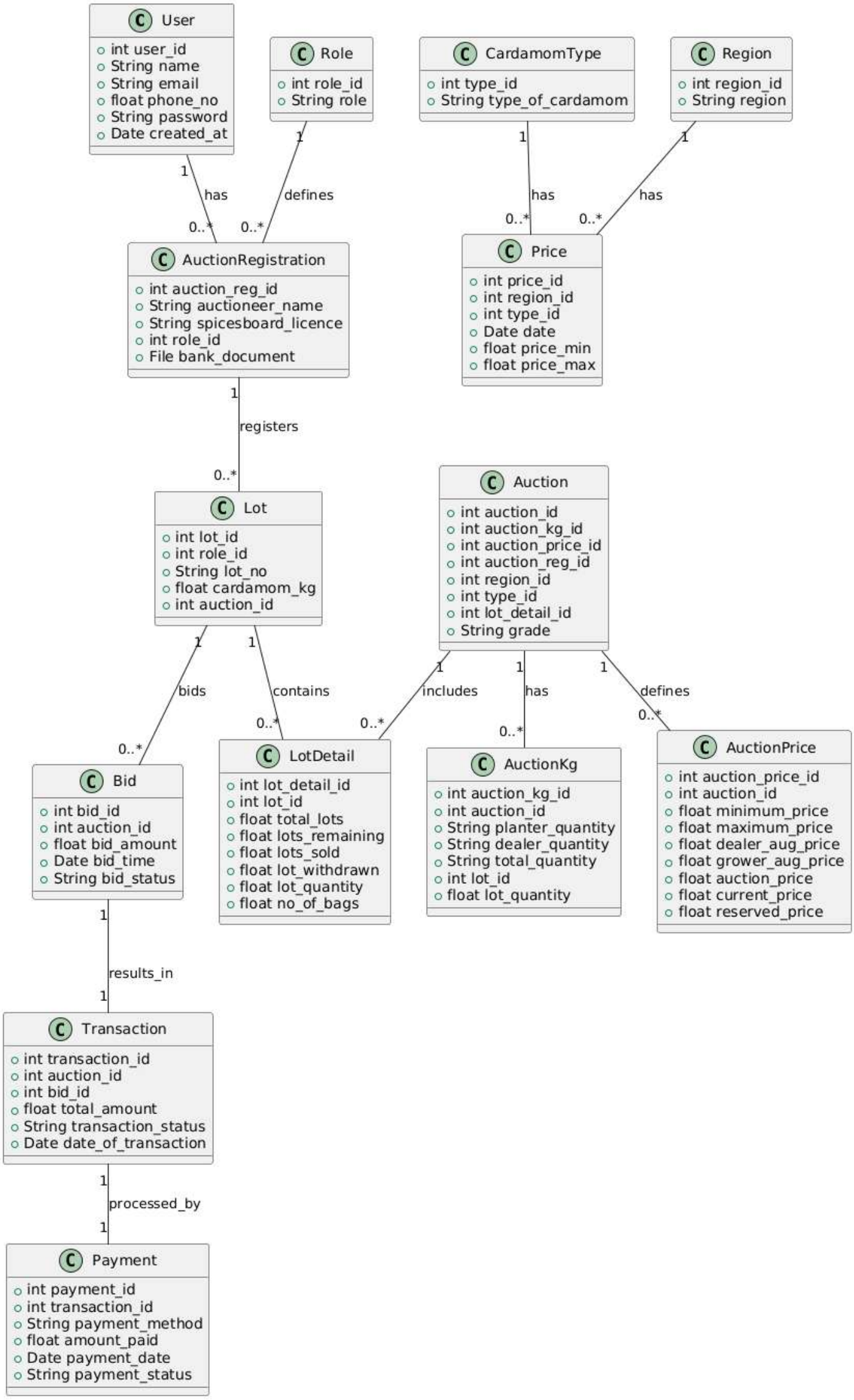
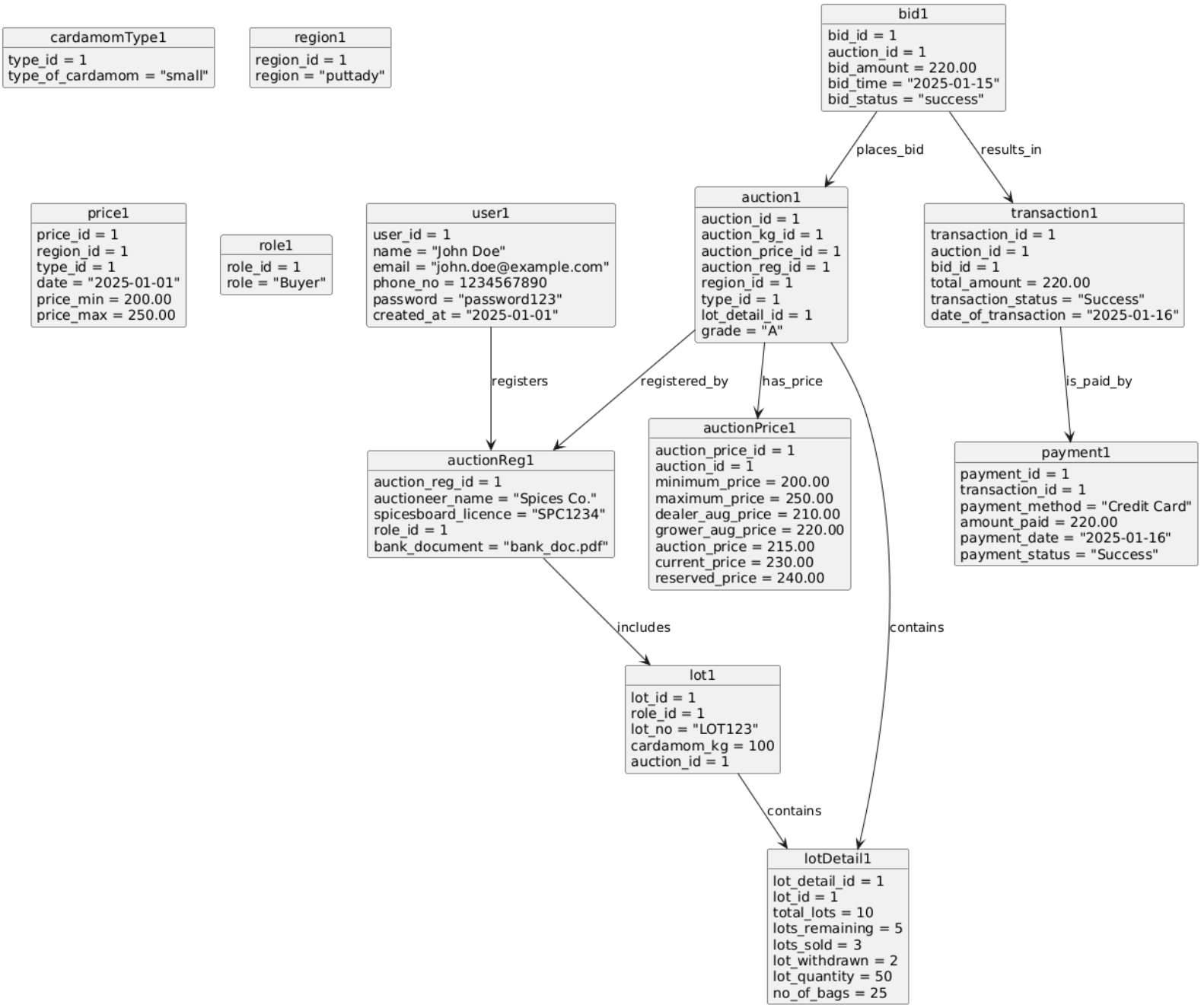
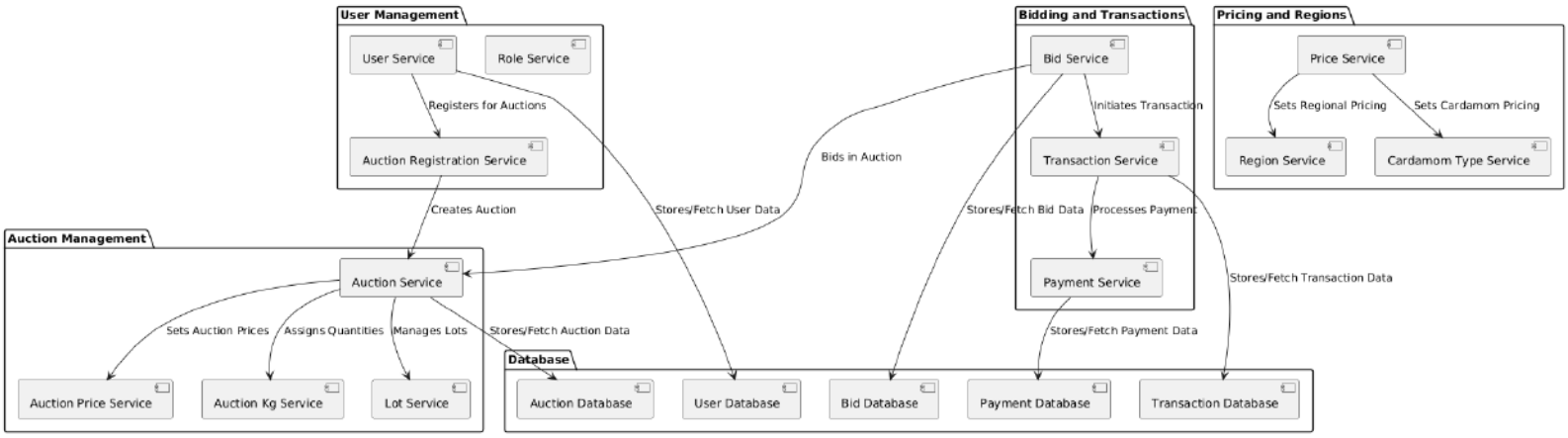
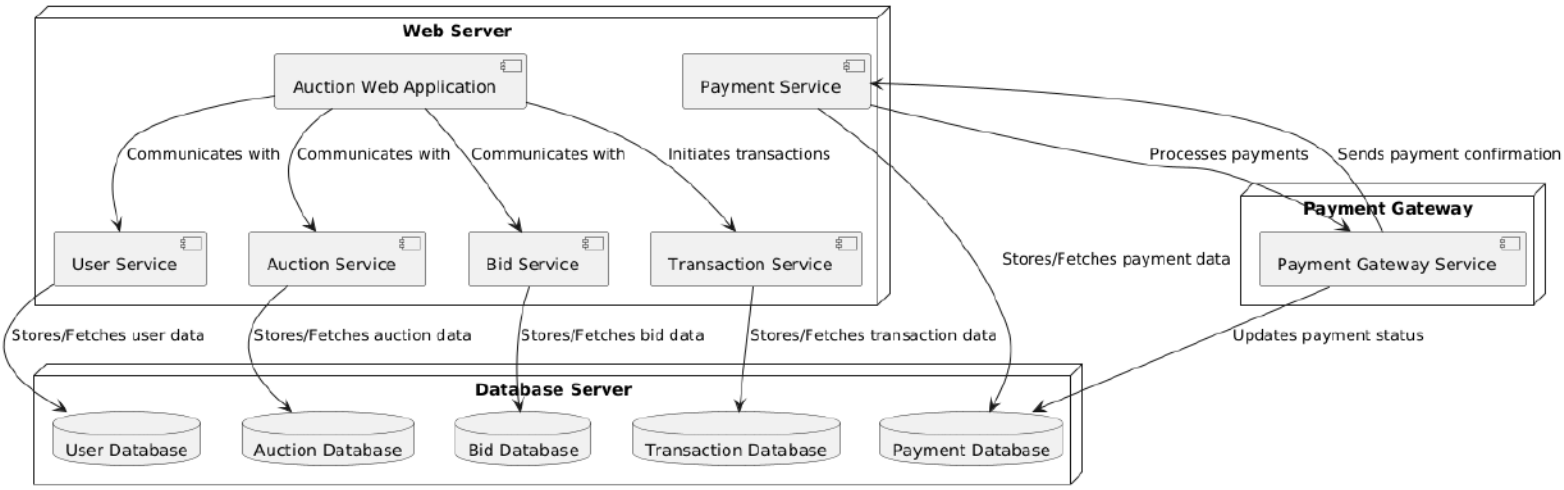
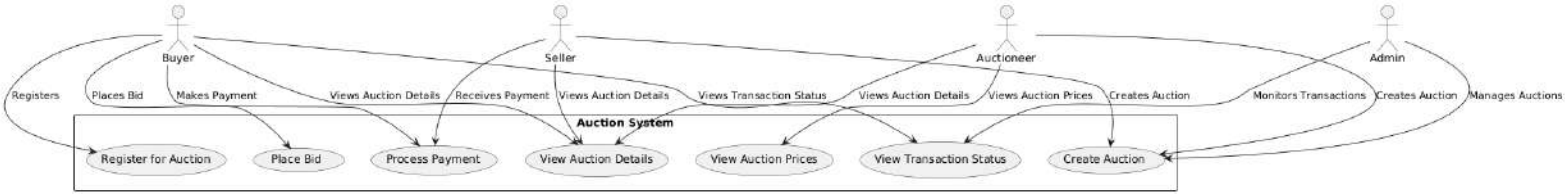
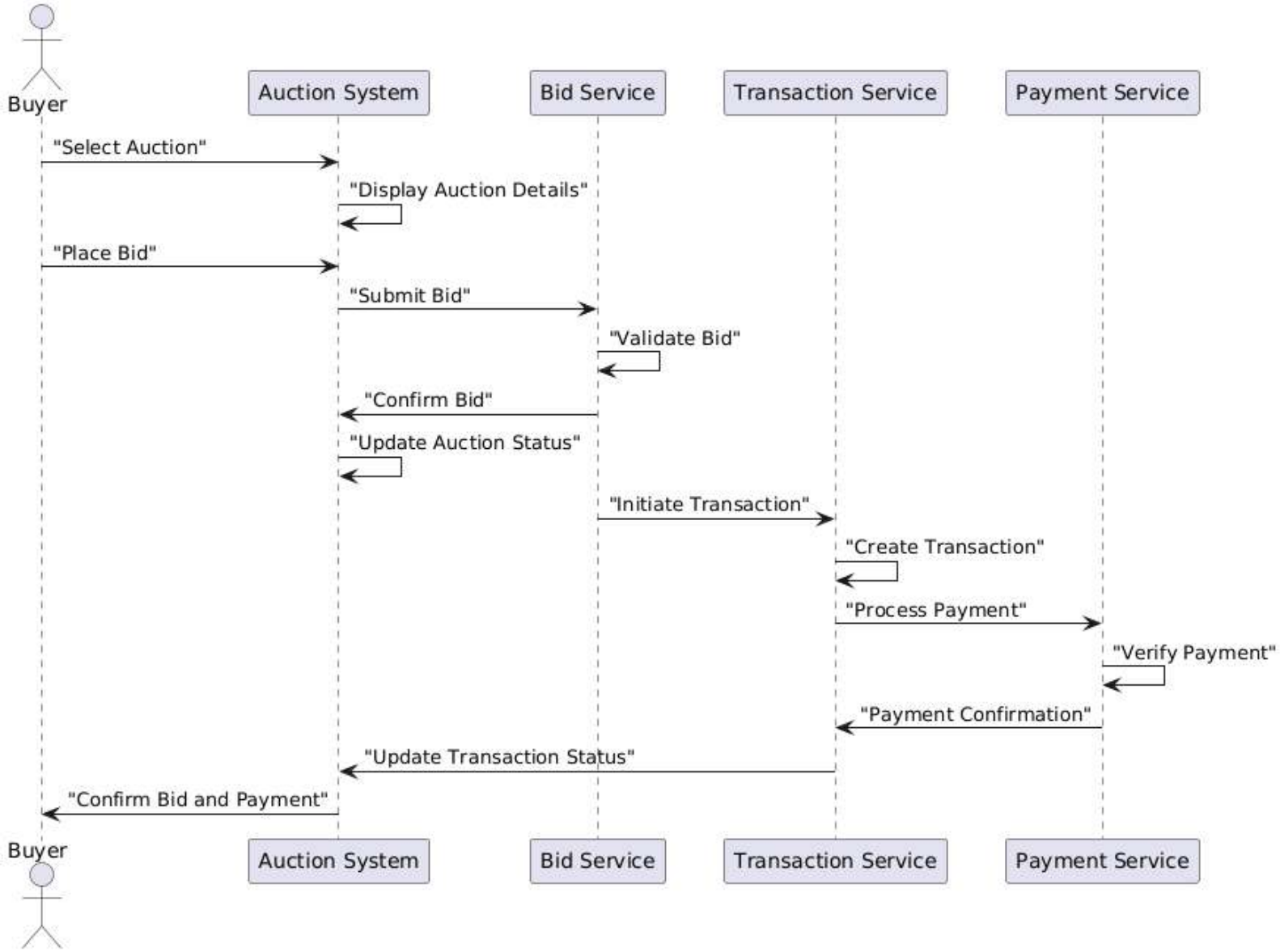

User
int user_id
String name
String email
float phone_no
String password
Date created_at
Role
int role_id
String role
CardamomType
int type_id
String type_of_cardamom
Region
int region_id
String region
1
has
0..*
1
defines
0..*
1
has
0..*
1
has
0..*
AuctionRegistration
int auction_reg_id
String auctioneer_name
String spicesboard_licence
int role_id
File bank_document
Price
int price_id
int region_id
int type_id
Date date
float price_min
float price_max
1
registers
0..*
Lot
int lot_id
int role_id
String lot_no
float cardamom_kg
int auction_id
Auction
int auction_id
int auction_kg_id
int auction_price_id
int auction_reg_id
int region_id
int type_id
int lot_detail_id
String grade
1
bids
0..*
1
contains
0..*
1
includes
0..*
1
has
0..*
1
defines
0..*
Bid
int bid_id
int auction_id
float bid_amount
Date bid_time
String bid_status
LotDetail
int lot_detail_id
int lot_id
float total_lots
float lots_remaining
float lots_sold
float lot_withdrawn
float lot_quantity
float no_of_bags
AuctionKg
int auction_kg_id
int auction_id
String planter_quantity
String dealer_quantity
String total_quantity
int lot_id
float lot_quantity
AuctionPrice
int auction_price_id
int auction_id
float minimum_price
float maximum_price
float dealer_aug_price
float grower_aug_price
float auction_price
float current_price
float reserved_price
1
results_in
1
Transaction
int transaction_id
int auction_id
int bid_id
float total_amount
String transaction_status
Date date_of_transaction
1
processed_by
1
Payment
int payment_id
int transaction_id
String payment_method
float amount_paid
Date payment_date
String payment_status

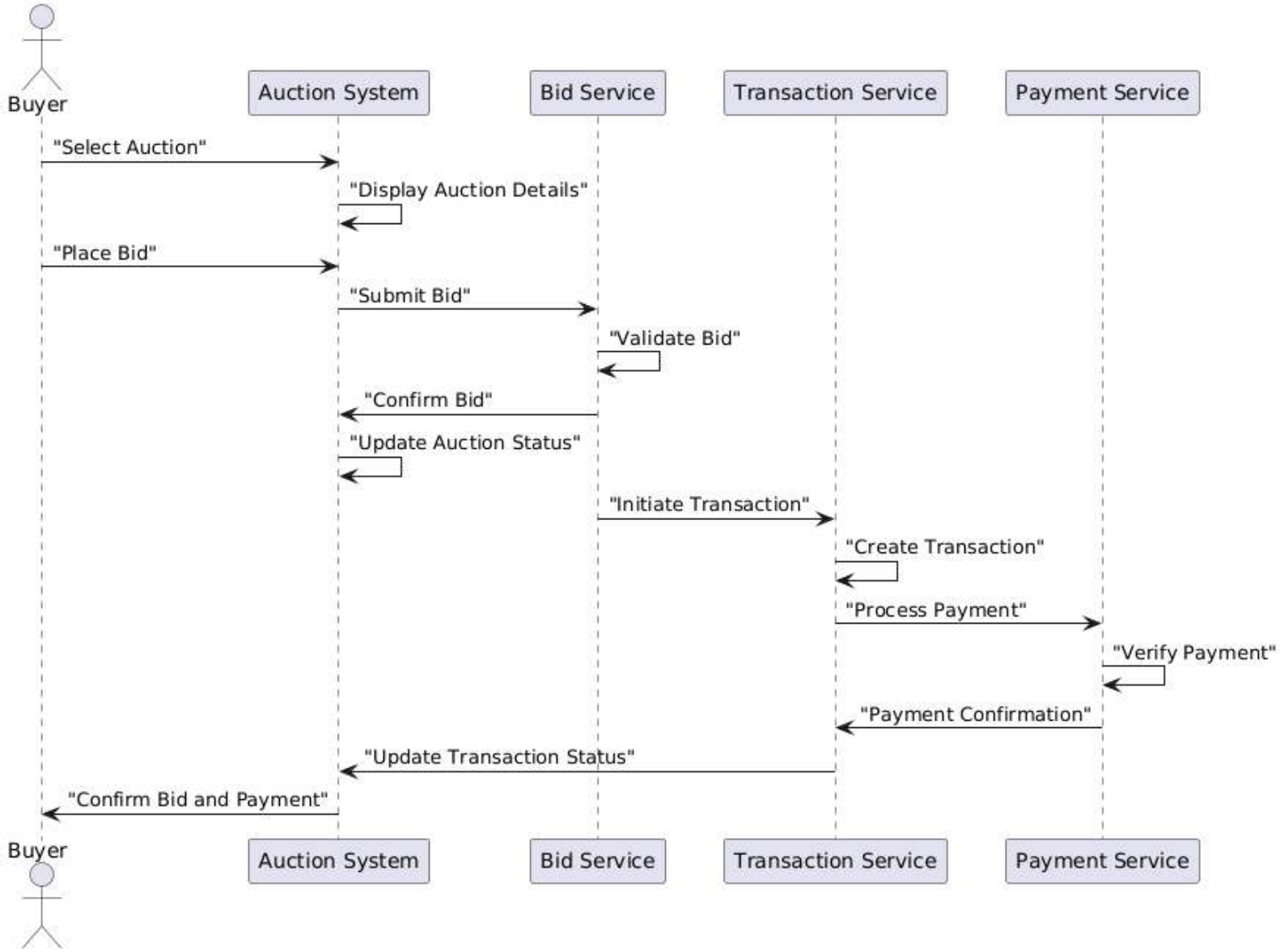

cardamomType1
type_id = 1
type_of_cardamom = "small"
region1
region_id = 1
region = "puttady"
bid1
bid_id = 1
auction_id = 1
bid_amount = 220.00
bid_time = "2025-01-15"
bid_status = "success"
places_bid
results_in
price1
price_id = 1
region_id = 1
type_id = 1
date = "2025-01-01"
price_min = 200.00
price_max = 250.00
role1
role_id = 1
role = "Buyer"
user1
user_id = 1
name = "John Doe"
email = "john.doe@example.com"
phone_no = 1234567890
password = "password123"
created_at = "2025-01-01"
auction1
auction_id = 1
auction_kg_id = 1
auction_price_id = 1
auction_reg_id = 1
region_id = 1
type_id = 1
lot_detail_id = 1
grade = "A"
transaction1
transaction_id = 1
auction_id = 1
bid_id = 1
total_amount = 220.00
transaction_status = "Success"
date_of_transaction = "2025-01-16"
registers
registered_by
has_price
is_paid_by
auctionReg1
auction_reg_id = 1
auctioneer_name = "Spices Co."
spicesboard_licence = "SPC1234"
role_id = 1
bank_document = "bank_doc.pdf"
auctionPrice1
auction_price_id = 1
auction_id = 1
minimum_price = 200.00
maximum_price = 250.00
dealer_aug_price = 210.00
grower_aug_price = 220.00
auction_price = 215.00
current_price = 230.00
reserved_price = 240.00
payment1
payment_id = 1
transaction_id = 1
payment_method = "Credit Card"
amount_paid = 220.00
payment_date = "2025-01-16"
payment_status = "Success"
includes
contains
lot1
lot_id = 1
role_id = 1
lot_no = "LOT123"
cardamom_kg = 100
auction_id = 1
contains
lotDetail1
lot_detail_id = 1
lot_id = 1
total_lots = 10
lots_remaining = 5
lots_sold = 3
lot_withdrawn = 2
lot_quantity = 50
no_of_bags = 25

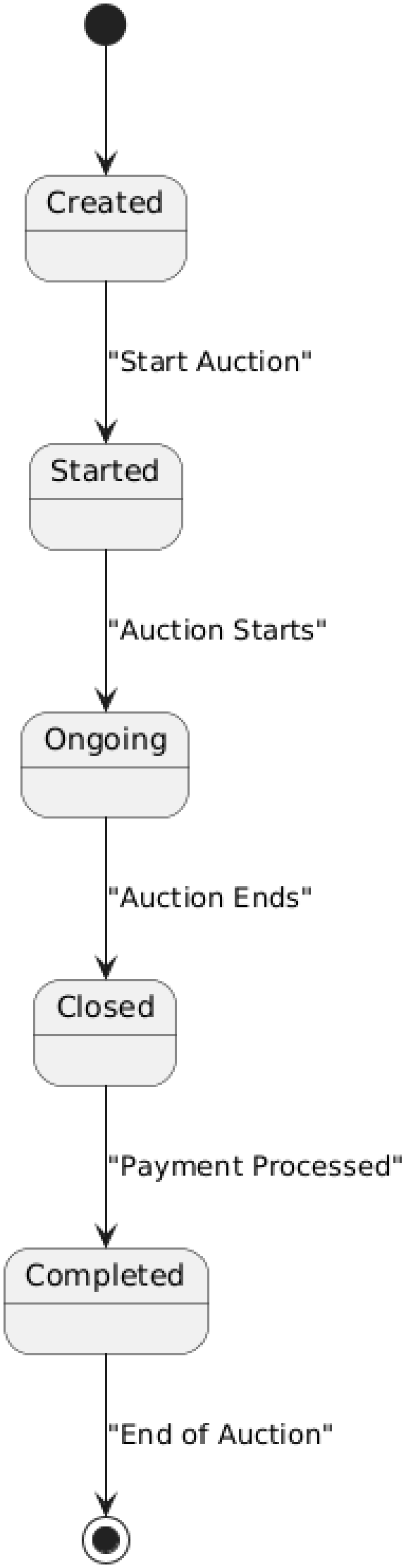

User Management
User Service
Role Service
Registers for Auctions
Auction Registration Service
Creates Auction
Stores/Fetch User Data
Bidding and Transactions
Bid Service
Initiates Transaction
Transaction Service
Processes Payment
Payment Service
Bids in Auction
Stores/Fetch Bid Data
Stores/Fetch Transaction Data
Stores/Fetch Payment Data
Pricing and Regions
Price Service
Sets Regional Pricing
Sets Cardamom Pricing
Region Service
Cardamom Type Service
Auction Management
Auction Service
Sets Auction Prices
Assigns Quantities
Manages Lots
Stores/Fetch Auction Data
Auction Price Service
Auction Kg Service
Lot Service
Database
Auction Database
User Database
Bid Database
Payment Database
Transaction Database

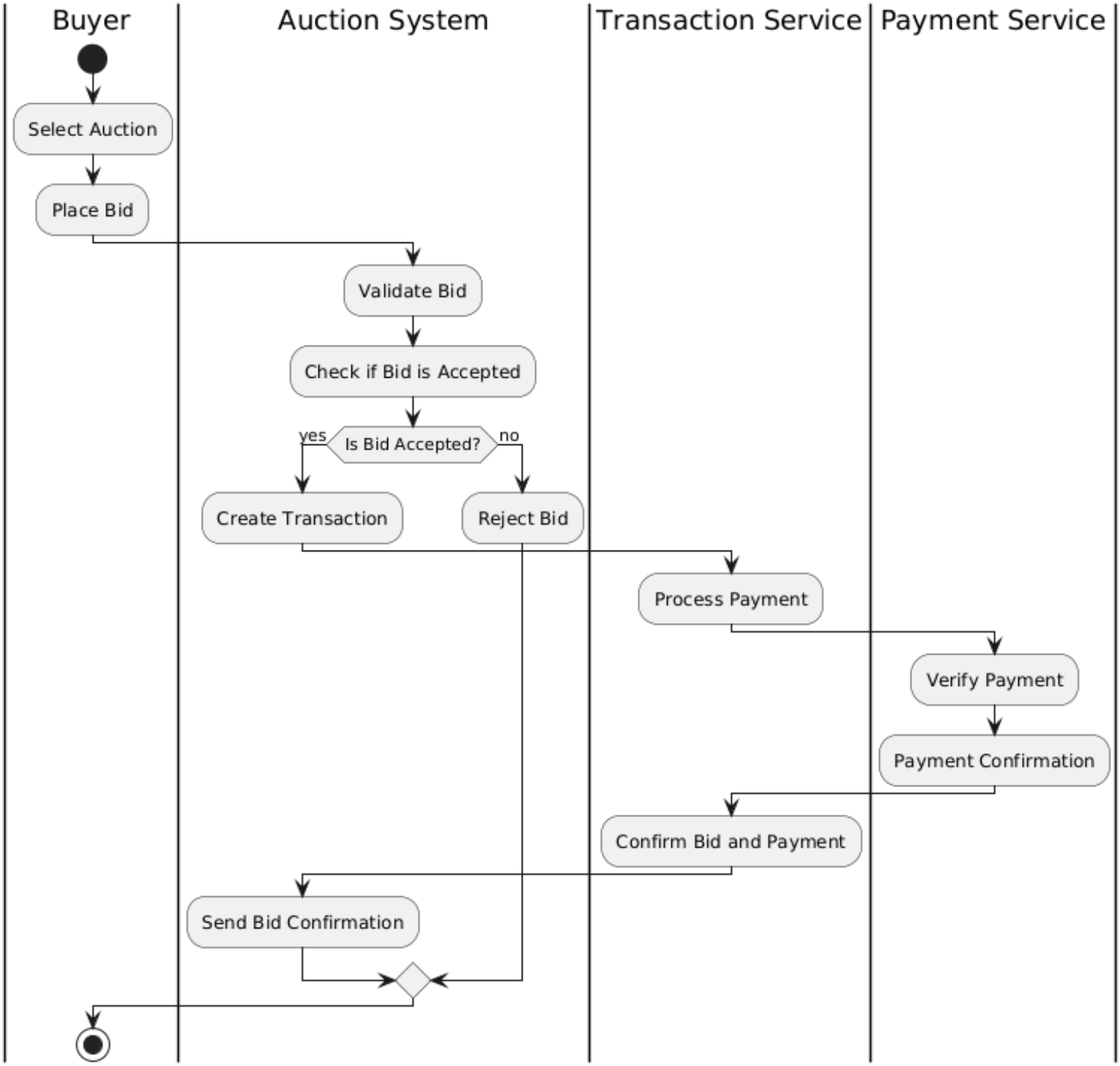

Web Server
Auction Web Application
Payment Service
Communicates with
Communicates with
Communicates with
Initiates transactions
Processes payments
Sends payment confirmation
Payment Gateway
Payment Gateway Service
Stores/Fetches payment data
User Service
Auction Service
Bid Service
Transaction Service
Stores/Fetches user data
Stores/Fetches auction data
Stores/Fetches bid data
Stores/Fetches transaction data
Updates payment status
Database Server
User Database
Auction Database
Bid Database
Transaction Database
Payment Database


Buyer
Seller
Auctioneer
Admin
Registers
Places Bid
Makes Payment
Views Auction Details
Receives Payment
Views Auction Details
Views Transaction Status
Views Auction Details
Views Auction Prices
Creates Auction
Monitors Transactions
Creates Auction
Manages Auctions
Auction System
Register for Auction
Place Bid
Process Payment
View Auction Details
View Auction Prices
View Transaction Status
Create Auction


Buyer
Auction System
Bid Service
Transaction Service
Payment Service
"Select Auction"
"Display Auction Details"
"Place Bid"
"Submit Bid"
"Validate Bid"
"Confirm Bid"
"Update Auction Status"
"Initiate Transaction"
"Create Transaction"
"Process Payment"
"Verify Payment"
"Payment Confirmation"
"Update Transaction Status"
"Confirm Bid and Payment"
Buyer
Auction System
Bid Service
Transaction Service
Payment Service


Buyer
Auction System
Bid Service
Transaction Service
Payment Service
"Select Auction"
"Display Auction Details"
"Place Bid"
"Submit Bid"
"Validate Bid"
"Confirm Bid"
"Update Auction Status"
"Initiate Transaction"
"Create Transaction"
"Process Payment"
"Verify Payment"
"Payment Confirmation"
"Update Transaction Status"
"Confirm Bid and Payment"
Buyer
Auction System
Bid Service
Transaction Service
Payment Service


Created
"Start Auction"
Started
"Auction Starts"
Ongoing
"Auction Ends"
Closed
"Payment Processed"
Completed
"End of Auction"


Buyer
Auction System
Transaction Service
Payment Service
Select Auction
Place Bid
Validate Bid
Check if Bid is Accepted
Is Bid Accepted?
yes
no
Create Transaction
Reject Bid
Process Payment
Verify Payment
Payment Confirmation
Confirm Bid and Payment
Send Bid Confirmation